

RATIONAL ACTIVATIONS AND LOSS-INFORMED OPTIMIZATION FOR TRANSFORMERS

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ABSTRACT

1 INTRODUCTION

2 RELATED WORK

3 METHOD

We introduce a learned activation and an optimizer for Transformer feed-forward blocks. GRAIN (Groupwise Rational Activation with Internal Normalization) learns a separate nonlinear response for each feature group. TILLER (Tangents Informed by a Loss Ledger for Equal-budget Reweighting) uses these responses to guide matrix updates and coordinate them across the network. The activation’s local geometry maps parameter directions to feature changes and their contributions to the loss.

3.1 GRAIN: LEARNING GROUPWISE ACTIVATION RESPONSES

The distribution and scale of intermediate features change during training. GRAIN evaluates a learned rational function in normalized coordinates, then restores the input scale. For a feed-forward input $x \in \mathbb{R}^d$, split the intermediate features into G groups of width w , with incoming and outgoing projections $A_g \in \mathbb{R}^{w \times d}$ and $B_g \in \mathbb{R}^{d \times w}$. The block output is

$$y = \sum_{g=1}^G B_g h_g, \quad h_g = \rho_g \phi_g(u_g), \quad u_g = \frac{A_g x}{\rho_g}, \quad \rho_g = \sqrt{\|A_g x\|_2^2 / w + \epsilon_{\text{act}}}. \quad (1)$$

Here ρ_g is the stabilized group root mean square (RMS), u_g contains the normalized features, and ϕ_g acts elementwise:

$$\phi_g(u) = \frac{\sum_{j=0}^5 a_{gj} u^j}{1 + \sum_{j=1}^4 |b_{gj}| |u|^j}. \quad (2)$$

The learned coefficients a_{gj}, b_{gj} control the response shape (Molina et al., 2020; Boullé et al., 2020); the denominator is at least one. Internal RMS normalization (Zhang & Sennrich, 2019) makes relative feature magnitudes the arguments of this function. Multiplication by ρ_g then carries the group’s amplitude into its output. Every group starts from the same SiLU approximation and learns its own response. Setting $Gw = 3H/2$ gives the two projections the same dense weight count as a SwiGLU block with intermediate width H (Shazeer, 2020).

GRAIN also provides a direct description of how an update changes its features. Let $J_g = \partial h_g / \partial (A_g x)$. An incoming direction q_g^A changes the features by $J_g q_g^A x$, whereas an outgoing direction q_g^B acts on the features as $q_g^B h_g$. The Jacobian contains the elementwise derivative and a correction for the change in group scale:

$$J_g = \text{Diag}(\phi'_g(u_g)) + \frac{1}{w} [\phi_g(u_g) - u_g \odot \phi'_g(u_g)] u_g^\top, \quad (3)$$

where $\odot$ denotes elementwise multiplication. These two response paths, through J_g and through h_g , provide the information used by TILLER. Appendix A derives the Jacobian.

3.2 TILLER: CONSTRUCTING AND COORDINATING LOSS-INFORMED UPDATES

TILLER first chooses a direction within each matrix block, then chooses how strongly the blocks should act together. Index the $C = LG$ layer-group pairs by $i = (\ell, g)$, where L is the Transformer depth. For either matrix role, let g_i be its clipped loss gradient, p_i its Muon direction, and a_i a direction formed by rescaling the same momentum with factorized gradient second moments (Jordan et al., 2024; Liu et al., 2025; Shazeer & Stern, 2018). The component $a_i^\perp = a_i - \langle a_i, p_i \rangle p_i / \|p_i\|_F^2$ is orthogonal to the Muon direction. It supplies a tangent along which TILLER can rotate that direction at fixed Frobenius norm.

The rotation depends on how the learned activation currently responds. A participation statistic π_i measures how broadly response energy is distributed: across singular directions of J_i for the incoming matrix, and across feature coordinates of h_i for the outgoing matrix. A congruence statistic κ_i measures alignment with the initial activation’s response, evaluated at the current inputs. Both statistics emphasize positions with greater downstream loss sensitivity. With $e_i = \pi_i(1 - \kappa_i^2)$, TILLER forms

$$\tilde{q}_i = \sqrt{1 - e_i} p_i + \sigma_i \sqrt{e_i} \frac{\|p_i\|_F}{\|a_i^\perp\|_F} a_i^\perp, \quad \sigma_i = \text{sign} \langle g_i, a_i^\perp \rangle. \quad (4)$$

Here $e_i \in [0, 1]$ is the fraction of squared direction norm assigned to the tangent, and $\text{sign}(0) = 1$ for its orientation. A broad response that has changed from initialization permits a larger rotation toward the adaptive direction. Orthogonality gives $\|\tilde{q}_i\|_F = \|p_i\|_F$. TILLER selects this candidate when it retains positive gradient contraction, with the precise selection rule and response statistics in Appendix B.1.

The selected incoming and outgoing directions, now denoted q_i^A, q_i^B , affect the loss jointly with directions in other groups. To measure these interactions, TILLER samples the same example-token positions across layers. At probe k , let $x_{k\ell}$ be the block input, h_{ki} its group features, and $\xi_{k\ell}$ the downstream loss gradient at the block output. The group’s scalar loss response is

$$S_{ki} = \langle \xi_{k\ell}, B_i J_{ki} q_i^A x_{k\ell} + q_i^B h_{ki} \rangle. \quad (5)$$

Each column records one group direction’s responses across probes. For group coefficients c , the vector Sc combines those responses; the cross terms in $\|Sc\|_2^2$ capture effects that reinforce or offset each other. Appendix B.2 derives this measurement and its relation to the minibatch gradient.

TILLER maintains a discounted second moment of these responses in a loss ledger, using Robust Frequent Directions to retain dominant response combinations and an isotropic spectral remainder (Ghashami et al., 2016; Luo et al., 2019). Let G_t be the resulting response metric, r_t its cross term with decoupled weight decay, and $s_{t,i} = \langle g_i^A, q_i^A \rangle + \langle g_i^B, q_i^B \rangle$ the current minibatch descent signal. At learning rate η , the coefficients are chosen using the quadratic surrogate

$$F_t(c) = -\eta s_t^\top c + \frac{\eta^2}{2} (c^\top G_t c + 2r_t^\top c), \quad \sum_i \omega_i c_i^2 = \sum_i \omega_i, \quad \omega_i = \|q_i^A\|_F^2 + \|q_i^B\|_F^2. \quad (6)$$

The linear term rewards current descent, while the quadratic term penalizes the magnitude of the combined loss response. The constraint preserves the total squared norm of the structured direction. Thus equal-budget reweighting allocates this norm among groups according to their joint loss effects. Signed coefficients can strengthen, attenuate, or reverse a group.

A projected solver proposes coefficients, which are evaluated in the full surrogate and checked for layerwise descent (Appendix B.4). The selected coefficients update $A_i \leftarrow (1 - \eta \lambda_{\text{wd}}) A_i - \eta c_i q_i^A$ and analogously for B_i , where λ_{wd} is weight decay. Attention uses the same response-guided rotation with layerwise statistics. Rational coefficients and the remaining parameter classes use AdamW (Loshchilov & Hutter, 2019). Appendices B.3 and C give the ledger, solver, and complete update.

4 EXPERIMENTS

5 DISCUSSION

6 CONCLUSION

AI USE STATEMENT

Generative AI tools assisted with code implementation, mathematical cross-checking, literature retrieval, and manuscript drafting. The authors reviewed the resulting material and take responsibility for the final content.

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A GRAIN: NORMALIZED RESPONSES AND THEIR DERIVATIVES

GRAIN learns an activation shape in coordinates defined by the input group’s own scale. This separates two aspects of the forward map: the rational function determines how relative feature values are transformed, and the group RMS carries their amplitude through the transformation. The derivative of this map describes the two paths through which the incoming and outgoing matrix directions affect the block output.

For one input $x \in \mathbb{R}^d$, the feed-forward block computes $z = Ax$, $h = \Phi(z)$, and $y = Bh$, where $A \in \mathbb{R}^{H_r \times d}$, $B \in \mathbb{R}^{d \times H_r}$, and $H_r = Gw$. A group consists of a row block A_g , the corresponding features $z_g = A_g x \in \mathbb{R}^w$, and a column block B_g . Its normalized input and output are

$$\rho_g = \sqrt{\|z_g\|_2^2 / w + \epsilon_{\text{act}}}, \quad u_g = z_g / \rho_g, \quad h_g = \Phi_g(z_g) = \rho_g \phi_g(u_g). \quad (7)$$

The block sums the group contributions as $y = \sum_g B_g h_g$. Each ϕ_g has the rational form in equation 2, applied elementwise with group-specific coefficients. Its denominator is at least one for every real input, so finite coefficients introduce no real poles. All groups begin from a common SiLU fit and subsequently learn independently. Appendix C specifies the initial coefficients.

An incoming update changes both u_g and ρ_g . Suppressing the group index, differentiation of $\rho = (z^\top z / w + \epsilon_{\text{act}})^{1/2}$ and $u = z / \rho$ gives

$$\nabla_z \rho = u / w, \quad \frac{\partial u}{\partial z} = \rho^{-1} \left(I - \frac{uu^\top}{w} \right). \quad (8)$$

Let $f = \phi(u)$, $d = \phi'(u)$, and $r = f - u \odot d$. The product rule for $\Phi(z) = \rho \phi(u)$ now gives

$$\begin{aligned} J &= \frac{\partial \Phi}{\partial z} = \frac{fu^\top}{w} + \text{Diag}(d) \left(I - \frac{uu^\top}{w} \right) \\ &= \text{Diag}(d) + \frac{ru^\top}{w}. \end{aligned} \quad (9)$$

These identities hold wherever ϕ is differentiable. At zero coordinates, the absolute-value terms use the automatic-differentiation selection $\text{sign}(0) = 0$; J denotes that selected derivative there.

The diagonal term applies the local slope separately to each feature. The second term accounts for their shared change in scale. For a linear response $\phi(u) = au$, $r = 0$ and $J = aI$: the normalization and scale restoration cancel. For a nonlinear response, r records the difference between the function value and its local linear contribution. Thus the same learned shape determines both the transformed features and their sensitivity to a perturbation.

For a feature perturbation v and a downstream derivative b , the two useful forms of this Jacobian are

$$Jv = d \odot v + r \frac{u^\top v}{w}, \quad J^\top b = d \odot b + u \frac{r^\top b}{w}. \quad (10)$$

The first maps an incoming perturbation to a feature change; the second maps loss sensitivity back through the activation. An update to B_g instead acts directly on h_g . This distinction explains why TILLER uses Jacobian geometry for incoming directions and feature geometry for outgoing directions.

B TILLER: FROM ACTIVATION GEOMETRY TO COORDINATED UPDATES

TILLER uses the activation response at two levels. Within a matrix block, the response determines a rotation between a Muon direction and a direction with adaptive coordinate scaling. Across blocks, loss responses determine how those directions should combine. The first construction supplies a direction for each group; the second assigns coefficients to the resulting collection.

B.1 CONSTRUCTING RESPONSE-GUIDED DIRECTIONS

Index the layer-group pairs by $i = (\ell, g)$ and the two projection roles by $r \in \{A, B\}$. For each role, let g_i^r be the current clipped gradient and m_i^r its Nesterov momentum. We construct a Muon

direction p_i^r from the approximate polar factor of this momentum, and an adaptive direction a_i^r from the same momentum divided by factorized gradient second moments (Jordan et al., 2024; Bernstein & Newhouse, 2024; Liu et al., 2025; Shazeer & Stern, 2018). The two directions retain different information: the polar map rebalances singular values, while the adaptive scaling reflects the history of individual row and column gradient magnitudes. Their numerical definitions are given in Appendix C.

The adaptive direction can change the orientation of p_i^r through its orthogonal component. Suppressing the role index, define

$$a_i^\perp = a_i - \frac{\langle a_i, p_i \rangle}{\|p_i\|_F^2} p_i, \quad \|p_i\|_F > 0, \quad (11)$$

where $\langle \cdot, \cdot \rangle$ is the Frobenius inner product for matrices. This tangent determines a plane of possible rotations. We choose the rotation magnitude from two complementary properties of the learned activation: the breadth of its response and its departure from the initial response.

Both properties are measured on shared example–token probes. For probe k , write h_{ki} and J_{ki} for the group features and activation Jacobian, and let $\xi_{k\ell} \in \mathbb{R}^d$ be the downstream loss derivative at the block output. An incoming update acts through J_{ki} , whose squared singular values describe its response energy. An outgoing update acts on h_{ki} , whose squared coordinates describe the available feature energy. The corresponding participation ratios are

$$\text{pr}_A(J) = \frac{[\text{tr}(JJ^\top)]^2}{w \|JJ^\top\|_F^2}, \quad \text{pr}_B(h) = \frac{\|h\|_2^4}{w \|h\|_4^4}. \quad (12)$$

Each ratio equals $1/w$ when all energy lies in one component and one when it is uniform across the w components. For example, if exactly m singular directions carry equal energy, $\text{pr}_A = m/w$. Participation therefore expresses response breadth on a common scale for the two matrix roles.

We weight these measurements by the loss sensitivity of the corresponding path:

$$\alpha_{ki}^A = \|J_{ki}^\top B_i^\top \xi_{k\ell}\|_2^2 / w, \quad \alpha_{ki}^B = \|\xi_{k\ell}\|_2^2 / d. \quad (13)$$

The weighted participation is

$$\pi_i^A = \frac{\sum_k \alpha_{ki}^A \text{pr}_A(J_{ki})}{\sum_k \alpha_{ki}^A}, \quad \pi_i^B = \frac{\sum_k \alpha_{ki}^B \text{pr}_B(h_{ki})}{\sum_k \alpha_{ki}^B}. \quad (14)$$

Consequently, the rotation reflects the response at positions where the loss is sensitive to changes in that group.

To measure learned change, compare the current response with the initial rational function evaluated at the same current input. Define response kernels $\Gamma_{ki}^A = J_{ki} J_{ki}^\top$ and $\Gamma_{ki}^B = h_{ki} h_{ki}^\top$, and let $\Gamma_{ki}^{r,0}$ use the initial rational coefficients. All other quantities, including the input and its RMS scale, remain at their current values. Their weighted congruence is

$$\kappa_i^r = \frac{\sum_k \alpha_{ki}^r \langle \Gamma_{ki}^r, \Gamma_{ki}^{r,0} \rangle}{\sqrt{\left(\sum_k \alpha_{ki}^r \|\Gamma_{ki}^r\|_F^2 \right) \left(\sum_k \alpha_{ki}^r \|\Gamma_{ki}^{r,0}\|_F^2 \right)}}. \quad (15)$$

This is a cosine between collections of weighted response kernels. It equals one when the two collections agree up to a common positive scale. Comparing them at the same input ties this measurement to the learned activation shape.

Participation and congruence together set the fraction of direction energy assigned to the adaptive tangent:

$$e_i^r = \pi_i^r (1 - (\kappa_i^r)^2). \quad (16)$$

A response spread over many components permits a larger rotation when its geometry has changed from initialization. A concentrated response reduces that rotation through π_i^r ; an unchanged response gives $e_i^r = 0$ through κ_i^r . Both factors are evaluated separately for incoming and outgoing directions.

The bounds needed for this construction follow directly from the definitions. For any nonnegative values v_j , $\sum_j v_j^2 \leq (\sum_j v_j)^2 \leq w \sum_j v_j^2$. Taking v_j to be squared singular values of J , or

squared coordinates of h , gives $\text{pr}_A, \text{pr}_B \in [1/w, 1]$ for nonzero responses. Congruence is at most one by Cauchy–Schwarz applied to the weighted kernels. Its numerator is nonnegative because $\langle J J^\top, J^0 J^{0\top} \rangle = \|J^\top J^0\|_F^2$ and $\langle h h^\top, h^0 h^{0\top} \rangle = (h^\top h^0)^2$. Thus $0 \leq e_i^r \leq 1$. Zero-response conventions appear in Appendix C.

We can now form the rotated direction. With role indices suppressed again, choose $\sigma_i = 1$ when $\langle g_i, a_i^\perp \rangle \geq 0$ and -1 otherwise, and set

$$\tilde{q}_i = \sqrt{1 - e_i} p_i + \sigma_i \sqrt{e_i} \frac{\|p_i\|_F}{\|a_i^\perp\|_F} a_i^\perp. \quad (17)$$

The sign makes the tangent contribution agree with current descent. Writing $e_i = \sin^2 \vartheta_i$ reveals the construction as a rotation through angle ϑ_i in the plane of p_i and $a_i^\perp$.

Proposition 1 (Norm-preserving rotation). *For $e_i \in [0, 1]$, $p_i \neq 0$, and $a_i^\perp \neq 0$, the candidate in equation 17 satisfies $\|\tilde{q}_i\|_F = \|p_i\|_F$.*

Proof. Equation 11 gives $\langle p_i, a_i^\perp \rangle = 0$. Expanding the squared norm therefore yields

$$\|\tilde{q}_i\|_F^2 = (1 - e_i) \|p_i\|_F^2 + e_i \frac{\|p_i\|_F^2}{\|a_i^\perp\|_F^2} \|a_i^\perp\|_F^2 = \|p_i\|_F^2.$$

□

The selected direction q_i is $\tilde{q}_i$ when the candidate is finite and both $\langle g_i, p_i \rangle > 0$ and $\langle g_i, \tilde{q}_i \rangle > 0$; otherwise it is p_i . A zero parent or tangent also selects p_i . Either selection preserves the parent norm. The resulting pair (q_i^A, q_i^B) now defines one structured direction whose strength can be coordinated with the other groups.

B.2 A JOINT LOSS MODEL FOR GROUP UPDATES

The directions constructed above occupy disjoint parameter blocks, but their effects on the loss overlap. A coefficient chosen for one group therefore changes the context in which the other groups act. TILLER measures this interaction by evaluating all group directions at shared activation positions.

For probe k and coordinate $i = (\ell, g)$, let $x_{k\ell}$ be the block input, $h_{ki} = \Phi_i(A_i x_{k\ell})$ the group features, and $\xi_{k\ell}$ the downstream loss derivative at the block output. The derivative is expressed on the clipped summed-loss scale specified in Appendix C. Perturbing the incoming and outgoing projections along (q_i^A, q_i^B) produces

$$\delta z_{ki} = q_i^A x_{k\ell}, \quad \delta h_{ki} = J_{ki} \delta z_{ki}, \quad \delta y_{ki} = B_i \delta h_{ki} + q_i^B h_{ki}. \quad (18)$$

The last expression follows from differentiating $(B_i + \alpha q_i^B) \Phi_i((A_i + \alpha q_i^A) x_{k\ell})$ at $\alpha = 0$. Its first term follows the incoming perturbation through the learned activation; its second term changes the outgoing projection of the current features. Pairing both terms with $\xi_{k\ell}$ gives the group’s loss response:

$$\begin{aligned} S_{t,ki} &= \langle \xi_{k\ell}, B_i J_{ki} q_i^A x_{k\ell} + q_i^B h_{ki} \rangle \\ &= \langle q_i^A x_{k\ell}, J_{ki}^\top B_i^\top \xi_{k\ell} \rangle + \langle q_i^B h_{ki}, \xi_{k\ell} \rangle. \end{aligned} \quad (19)$$

The second form uses the Jacobian action in equation 10. At differentiability points, each score is the contribution from one activation position to the directional loss derivative. Averaging over all positions recovers the clipped minibatch derivative. The probe matrix $S_t \in \mathbb{R}^{K \times C}$, where $C = LG$, retains K such positions across all group directions. The exact current minibatch derivative for each coordinate is also available from the parameter gradients:

$$s_{t,i} = \langle g_{t,i}^A, q_{t,i}^A \rangle + \langle g_{t,i}^B, q_{t,i}^B \rangle. \quad (20)$$

This contraction supplies the linear descent term, while the probe rows describe how contributions vary together.

For coefficients $c \in \mathbb{R}^C$, the combined probe response is $S_t c$. The quantity $\|S_t c\|_2^2 / K$ depends on cross products between its columns. To illustrate what these products reveal, consider two probes

and two group responses. Columns $(2, 0)^\top$ and $(0, 2)^\top$ each have mean one, and their sum has mean squared response four. If both columns are $(2, 0)^\top$, the individual means are unchanged but the combined mean squared response is eight. The response metric distinguishes these combinations through their overlap across probes.

Decoupled weight decay contributes to the same output perturbation. Applying equation 19 to $(\lambda_{\text{wd}} A_i, \lambda_{\text{wd}} B_i)$ and summing over groups gives its probe response $a_t^{\text{dec}} \in \mathbb{R}^K$. The total response penalty expands as

$$\frac{1}{K} \|S_t c + a_t^{\text{dec}}\|_2^2 = c^\top \frac{S_t^\top S_t}{K} c + 2c^\top \frac{S_t^\top a_t^{\text{dec}}}{K} + \frac{\|a_t^{\text{dec}}\|_2^2}{K}. \quad (21)$$

The last term is constant in c . The first two terms motivate a response metric G_t and a score-decay cross term r_t . TILLER estimates them from a discounted history of measurements, as described in the next subsection.

The coordination problem balances current descent against this joint response penalty. Let $\omega_i = \|q_i^A\|_F^2 + \|q_i^B\|_F^2$ and $\mathcal{E}_0 = \sum_i \omega_i$. Because the coordinates occupy disjoint parameter blocks, their structured update satisfies

$$\|\Delta\theta(c)\|_2^2 = \eta^2 \sum_i \omega_i c_i^2, \quad \Delta\theta(c) = -\eta \{c_i q_i^A, c_i q_i^B\}_{i=1}^C.$$

Thus $\sum_i \omega_i c_i^2 = \mathcal{E}_0$ preserves the squared norm of the uncoordinated direction $c = \mathbf{1}$. At learning rate $\eta > 0$, TILLER uses the surrogate

$$F_t(c) = -\eta s_t^\top c + \frac{\eta^2}{2} (c^\top G_t c + 2r_t^\top c), \quad (22)$$

$$\text{with } \sum_i \omega_i c_i^2 = \mathcal{E}_0. \quad (23)$$

The linear term favors combinations aligned with the current minibatch gradient. The quadratic term penalizes their joint squared loss response, including interaction with decay. Fixing the norm makes the choice depend on how that norm is distributed among groups. Signed coefficients permit a group to reverse when that change benefits the joint proposal.

B.3 MAINTAINING THE LOSS LEDGER

The coefficient model needs both a current descent signal and a description of how directions interact. TILLER obtains the first from s_t on every minibatch. It obtains the second from shared probes and carries those measurements forward in a discounted loss ledger.

The shared probes are refreshed on step 1 and every eight steps thereafter, using $K = 32$ example-token positions. Between refreshes, the ledger receives a rank-one measurement aligned with the current minibatch contraction. Let $\nu = \|S_{\text{ref}}\|_F / \sqrt{K}$ be the RMS row norm at the latest probe refresh. Define

$$\gamma_t = \begin{cases} \nu / \|s_t\|_2, & \|s_t\|_2 > 0, \\ 0, & \|s_t\|_2 = 0, \end{cases} \quad S_t = \mathbf{1}_K (\gamma_t s_t)^\top, \quad a_t^{\text{dec}} = \mathbf{1}_K \gamma_t d_t^{\text{dec}}, \quad (24)$$

where d_t^{dec} is the current gradient contraction with the GRAIN decay direction. When $s_t \neq 0$, $\|S_t\|_F^2 = K \gamma_t^2 \|s_t\|_2^2 = \|S_{\text{ref}}\|_F^2$. This preserves the trace of the latest probe second moment while updating its orientation to the current descent signal. Probe refreshes restore variation across positions; intervening steps supply fresh information along s_t .

The ledger’s uncompressed target is

$$G_t^* = \beta_2 G_{t-1}^* + (1 - \beta_2) S_t^\top S_t / K, \quad G_1^* = S_1^\top S_1 / K. \quad (25)$$

For any c , $c^\top G_t^* c$ is a discounted average of squared responses of that coefficient combination. The recorded responses use the parameters and directions from their respective steps. The discount therefore gives greater weight to the geometry encountered more recently.

TILLER represents this history using a factor $\mathcal{B}_t$ and an isotropic remainder $\tau_t I$:

$$\widehat{G}_t = \mathcal{B}_t^\top \mathcal{B}_t / K + \tau_t I. \quad (26)$$

The factor represents dominant combinations of group responses. The remainder assigns a common contribution to all coefficient directions, including those outside the factor's span. Given the previous history, the current rows enter through

$$\mathcal{A}_t = \begin{bmatrix} \sqrt{\beta_2} \mathcal{B}_{t-1} \\ \sqrt{1 - \beta_2} S_t \end{bmatrix}, \quad G_t = \mathcal{A}_t^\top \mathcal{A}_t / K + \beta_2 \tau_{t-1} I. \quad (27)$$

This G_t supplies the current coefficient model. The cross term follows the same discount:

$$r_t = \beta_2 r_{t-1} + (1 - \beta_2) S_t^\top a_t^{\text{dec}} / K. \quad (28)$$

At the first step, $\mathcal{A}_1 = \mathcal{B}_1 = S_1$, $\tau_0 = \tau_1 = 0$, and $r_1 = S_1^\top a_1^{\text{dec}} / K$.

To carry the new history to the next step, TILLER compresses $\mathcal{A}_t$ with Robust Frequent Directions (Ghashami et al., 2016; Luo et al., 2019). It retains at most $r = 64$ singular directions and subtracts the largest discarded squared singular value Δ_t from each retained squared singular value. The isotropic remainder then updates as

$$\tau_t = \beta_2 \tau_{t-1} + \frac{\Delta_t}{2K}. \quad (29)$$

The new rows thus enter the current model at full resolution, and their compressed representation becomes part of the next step's history.

The half-threshold in equation 29 has a geometric interpretation. After shrinkage, the omitted contribution lies between zero and Δ_t / K in every direction. Assigning the midpoint of this interval limits the largest absolute error. The following result quantifies its effect on the response metric.

Proposition 2 (Error of the compressed response metric). *Let $K > 0$, $0 \leq \beta_2 \leq 1$, $\tau_{t-1} \geq 0$, and $\mathcal{A} = U \text{Diag}(\sigma) V^\top$, with singular values in nonincreasing order. If $\text{rank}(\mathcal{A}) > r$, set $\Delta = \sigma_{r+1}^2$ and*

$$\mathcal{B} = \text{Diag}\left(\sqrt{\sigma_1^2 - \Delta}, \dots, \sqrt{\sigma_r^2 - \Delta}\right) V_{1:r}^\top. \quad (30)$$

Otherwise set $\Delta = 0$ and choose a factor with at most r rows satisfying $\mathcal{B}^\top \mathcal{B} = \mathcal{A}^\top \mathcal{A}$. For

$$G = \mathcal{A}^\top \mathcal{A} / K + \beta_2 \tau_{t-1} I, \quad \widehat{G} = \mathcal{B}^\top \mathcal{B} / K + \left(\beta_2 \tau_{t-1} + \frac{\Delta}{2K}\right) I,$$

we have

$$\widehat{G} \succeq 0, \quad \|\widehat{G} - G\|_2 \leq \frac{\Delta}{2K}. \quad (31)$$

Proof. The retained squared singular values and the isotropic remainder are nonnegative, so $\widehat{G}$ is positive semidefinite. If $\Delta = 0$, the two metrics coincide. Otherwise, on a retained right-singular direction, shrinkage subtracts Δ / K and the remainder adds $\Delta / (2K)$, for a net error of $-\Delta / (2K)$. On a discarded direction with squared singular value $\lambda \in [0, \Delta]$, the error is $(\Delta / 2 - \lambda) / K$. A null direction has error $\Delta / (2K)$. These directions form an orthonormal basis, and every absolute eigenvalue error is at most $\Delta / (2K)$. $\square$

The bound translates directly to the penalty used for coordination. For any coefficient vector c ,

$$|c^\top \widehat{G} c - c^\top G c| \leq \|\widehat{G} - G\|_2 \|c\|_2^2 \leq \frac{\Delta}{2K} \|c\|_2^2. \quad (32)$$

It also tracks error over the discounted history. Writing $E_t = \widehat{G}_t - G_t^*$, each compression adds an error D_t with $\|D_t\|_2 \leq \Delta_t / (2K)$, so $E_t = \beta_2 E_{t-1} + D_t$ and $E_1 = 0$. Iteration yields

$$\|E_t\|_2 \leq \sum_{j=2}^t \beta_2^{t-j} \frac{\Delta_j}{2K}. \quad (33)$$

The same discount that emphasizes recent responses therefore attenuates older compression errors.

B.4 CHOOSING THE GROUP COEFFICIENTS

The ledger supplies G_t and r_t , while the current gradient supplies s_t . Coefficient selection combines these quantities in equation 22. A change of coordinates turns the weighted norm constraint into a sphere and reduces the projected solution to a scalar multiplier equation.

For $\omega_i > 0$, let $W = \text{Diag}(\omega)$ and $y = W^{1/2}c$. Substituting $c = W^{-1/2}y$ and dividing the objective by η^2 gives

$$\mathcal{Q}_t(y) = \frac{1}{2}y^\top M_t y - b_t^\top y, \quad \|y\|_2^2 = \mathcal{E}_0, \quad (34)$$

$$M_t = W^{-1/2}G_t W^{-1/2}, \quad b_t = W^{-1/2}(s_t/\eta - r_t).$$

The coefficient construction assumes $b_t \neq 0$, in addition to positive block energies and learning rate. Its Lagrangian is $\mathcal{J}(y, \lambda) = \frac{1}{2}y^\top M_t y - b_t^\top y + \frac{\lambda}{2}(\|y\|_2^2 - \mathcal{E}_0)$. Differentiation with respect to y gives the stationarity equation $(M_t + \lambda I)y = b_t$. The multiplier controls the norm of the proposed update in these coordinates.

TILLER constructs a proposal on the positive-multiplier branch in a subspace generated by b_t and repeated actions of M_t . A reorthogonalized Lanczos recurrence (Lanczos, 1950) produces a basis Q_k , with $k \leq 32$. In the absence of breakdown, its span is $\text{span}\{b_t, M_t b_t, \dots, M_t^{k-1} b_t\}$. Each metric action follows directly from the ledger:

$$M_t v = W^{-1/2} \mathcal{A}_t^\top (\mathcal{A}_t W^{-1/2} v) / K + \beta_2 \tau_{t-1} W^{-1} v. \quad (35)$$

Let T_k^+ be the positive-semidefinite projection of the accumulated Lanczos matrix. With $y = Q_k z$, the projected stationarity equation is

$$(T_k^+ + \lambda I)z = Q_k^\top b_t, \quad \|z\|_2^2 = \mathcal{E}_0. \quad (36)$$

Diagonalizing $T_k^+ = U \text{Diag}(\theta) U^\top$ and writing $\gamma = U^\top Q_k^\top b_t$ reduces the norm condition to

$$\psi(\lambda) = \sum_j \frac{\gamma_j^2}{(\theta_j + \lambda)^2} = \mathcal{E}_0. \quad (37)$$

For $\lambda > 0$, nonnegative θ_j and nonzero γ give $\psi'(\lambda) = -2 \sum_j \gamma_j^2 / (\theta_j + \lambda)^3 < 0$. Thus bisection determines the unique positive root when one lies in the chosen interval. The resulting proposal is $c = W^{-1/2} Q_k U \text{Diag}((\theta + \lambda)^{-1}) \gamma$.

The proposal is evaluated in the full objective F_t , using the ledger factor and isotropic remainder. It is selected when it is finite, lowers F_t relative to $c = \mathbf{1}$, and satisfies

$$\frac{|c^\top W c - \mathcal{E}_0|}{\max\{\mathcal{E}_0, 1\}} \leq 10^{-8}. \quad (38)$$

TILLER also requires positive aggregate contractions with the current gradient and Nesterov momentum in each layer and matrix role:

$$\sum_g c_{\ell g} \langle g_{\ell g}^r, q_{\ell g}^r \rangle > 0, \quad \sum_g c_{\ell g} \langle m_{\ell g}^r, q_{\ell g}^r \rangle > 0, \quad r \in \{A, B\}. \quad (39)$$

These inequalities are checked for both the proposed coefficients and $\mathbf{1}$. If the positive root is unavailable or a check fails, the step uses $c = \mathbf{1}$. The selected coefficients scale both projections of each group, completing the coordination of the response-guided directions.

C ALGORITHM SPECIFICATION

A training step combines the constructions above in the following order. The forward and backward passes provide the clipped parameter gradients and, on a probe refresh, the activation values and downstream loss derivatives. The optimizer then constructs directions, measures their joint response, and applies the selected coefficients.

1. Update Nesterov momentum and factorized second moments for the GRAIN projections and attention matrices, and construct their Muon and adaptive directions.

2. On a probe refresh, evaluate the participation and congruence statistics; otherwise reuse their latest values. Form the response-guided rotations and select the matrix directions.
3. Evaluate the shared probe scores and decay responses on a refresh step. At other steps, use the current gradient contractions and equation 24.
4. Combine the current rows with the previous ledger to form G_t and r_t . Construct and evaluate the coefficient proposal. Compress the response factor and retain the updated ledger for the next step.
5. Apply the paired GRAIN updates

$$A_i \leftarrow (1 - \eta\lambda_{\text{wd}})A_i - \eta c_i q_i^A, \quad B_i \leftarrow (1 - \eta\lambda_{\text{wd}})B_i - \eta c_i q_i^B.$$

Apply the response-guided directions to attention, and update the rational coefficients and remaining parameter classes with AdamW (Loshchilov & Hutter, 2019).

For the activation, the group width is $w = 256$ and $\epsilon_{\text{act}} = 10^{-6}$. The expanded width is $H_r = Gw$. All groups begin from the same SiLU fit. In ascending powers, the coefficients are

$$(a_0^{\text{init}}, \dots, a_5^{\text{init}}) = (2.4915714620520875 \times 10^{-5}, 0.5000668663485823, 0.2499441908995771, 0.0526200910151016, 0.005525765640115002, 0.00024199321178747251), \quad (40)$$

$$(b_1^{\text{init}}, \dots, b_4^{\text{init}}) = (0.00022554017910995938, 0.10511420350691994, 2.8656992394109812 \times 10^{-5}, 0.0004816962110562025). \quad (41)$$

The ten coefficients subsequently learn independently in every group.

The matrix directions use the same momentum in their polar and adaptive constructions. Let g_t denote the parameter gradient after microbatch accumulation, distributed averaging, and global-norm clipping. For each pair of group blocks $i = (\ell, g)$ and role $r \in \{A, B\}$, TILLER starts from a Nesterov–Muon direction p_i^r : with $\beta_1 = 0.95$, it forms

$$v_{t,i}^r = \beta_1 v_{t-1,i}^r + (1 - \beta_1) g_{t,i}^r, \quad m_{t,i}^r = \beta_1 v_{t,i}^r + (1 - \beta_1) g_{t,i}^r, \quad (42)$$

followed by a five-step Newton–Schulz polar map applied independently to each group block and Muon’s role-dependent RMS adjustment (Jordan et al., 2024; Bernstein & Newhouse, 2024; Liu et al., 2025). To set the groupwise polar directions on the nominal Frobenius scale of their full projection matrix, we apply

$$\chi_{\text{grp}} = \sqrt{\frac{\min\{H_r, d\}}{G \min\{w, d\}}}, \quad (43)$$

Writing $\mathcal{P}_5$ for this blockwise polar map, Muon’s adjustment is $\mu_{\text{mlp}} = 0.2\sqrt{\max\{H_r, d\}}$ for either GRAIN matrix role. The parent group direction is $p_{t,i}^r = \mu_{\text{mlp}} \chi_{\text{grp}} \mathcal{P}_5(m_{t,i}^r)$. In parallel, TILLER maintains factorized row and column second moments, in the spirit of memory-efficient factored adaptation (Shazeer & Stern, 2018). For a grouped matrix with gradient g_i and Nesterov momentum m_i , suppressing role indices,

$$R_t = \beta_2 R_{t-1} + (1 - \beta_2) \text{rowsum}(g_i^2), \quad (44)$$

$$C_t = \beta_2 C_{t-1} + (1 - \beta_2) \text{colsum}(g_i^2), \quad (45)$$

$$V_{t,jk} = \frac{R_{t,j} C_{t,k}}{\max\{\sum_s R_{t,s}, \delta_{\text{fp}}\}} \frac{1}{1 - \beta_2^t}, \quad a_i = \mu_{\text{mlp}} \frac{m_i}{\sqrt{V_t} + \epsilon_{\text{opt}}}. \quad (46)$$

Here rowsum and colsum sum squared entries over the columns and rows, respectively. We set $\beta_2 = 0.95$, $\epsilon_{\text{opt}} = 10^{-8}$, and $\delta_{\text{fp}} = \text{finfo}(\text{float32}).\text{tiny}$. The same β_2 sets the time scale of the loss ledger. When the row-moment total is zero, $V_t = 0$.

The approximate polar map $\mathcal{P}_5$ transposes a tall input and casts the working matrix to bfloat16. Its Frobenius norm is accumulated in float32, lower bounded by 10^{-7} , and cast to bfloat16 for normalization. With $(a, b, c) = (3.4445, -4.775, 2.0315)$, its five iterations are

$$X_{j+1} = aX_j + b(X_j X_j^\top)X_j + c(X_j X_j^\top)^2 X_j. \quad (47)$$

The iterations run in bfloat16, after which the original orientation is restored. Factorized second moments are maintained in float32.

The response statistics use deterministic shared probes at steps 1, 9, 17, All quantities in a probe row refer to the same example–token position. If the backward pass averages over R positions per microbatch and M accumulated microbatches, a captured output derivative $\bar{\xi}_{k\ell}$ is converted to

$$\xi_{k\ell} = RM\chi_{\text{clip}}\bar{\xi}_{k\ell}, \quad (48)$$

where χ_{clip} is the realized global gradient-clipping factor. This is the derivative of the summed microbatch loss, scaled by χ_{clip} . The clipping factor is held fixed when forming the directional measurements. For zero responses, $\text{pr}_A(0) = \text{pr}_B(0) = 0$. If either role has zero total sensitivity, both participation values are set to one. If either kernel-congruence denominator vanishes, both congruence values are set to one. Congruence is also set to one when the rational coefficients equal their initialization. These conventions select zero departure when the response comparison is undefined.

Attention uses the same rotation construction with statistics aggregated within its layer. Let $\bar{\pi}_\ell^r = G^{-1} \sum_g \pi_{\ell g}^r$. For each role, obtain κ_ℓ^r by summing the numerator and the two squared-norm sums in equation 15 over groups before normalization. Then

$$\pi_\ell^{\text{att}} = \sqrt{\bar{\pi}_\ell^A \bar{\pi}_\ell^B}, \quad \kappa_\ell^{\text{att}} = \sqrt{\kappa_\ell^A \kappa_\ell^B}, \quad e_\ell^{\text{att}} = \pi_\ell^{\text{att}} [1 - (\kappa_\ell^{\text{att}})^2]. \quad (49)$$

The participation term combines the breadth of both projection roles, while the congruence term allows a change in either response to affect the attention rotation. This fraction guides the QKV and attention-output matrices. The 16 attention heads are partitioned into blocks of four for the polar map; the stacked QKV direction is scaled by $1/\sqrt{3}$ for its three constituent projections. The adaptive tangent uses matrix-level factorized moments. The group coefficients c_i apply to the paired GRAIN projections; attention receives its layerwise rotation. Rational coefficients, embeddings, normalization parameters, and remaining tensors use AdamW.

The coefficient solver uses double precision. Its Lanczos recurrence performs two reorthogonalization passes per basis vector. At an early breakdown, a deterministic cosine vector is reorthogonalized against the existing basis to start a new block, with zero coupling between blocks. The basis is checked for orthogonality before use. If $\bar{T}_k = U \text{Diag}(\bar{\theta}) U^\top$ is the accumulated projected matrix, the solver sets $T_k^+ = U \text{Diag}((\bar{\theta})_+) U^\top$. The positive multiplier is bracketed by

$$\lambda_{\text{lo}} = \epsilon_{\text{mach}} C, \quad \lambda_{\text{hi}} = \frac{\|b_t\|_2}{\sqrt{\mathcal{E}_0}} + \lambda_{\text{max}}(T_k^+) + \lambda_{\text{lo}}. \quad (50)$$

Since $\theta_j \geq 0$, $\psi(\lambda_{\text{hi}}) \leq \|b_t\|_2^2 / \lambda_{\text{hi}}^2 \leq \mathcal{E}_0$. When $\psi(\lambda_{\text{lo}}) \geq \mathcal{E}_0$, the interval contains the positive root and the solver takes 64 bisection steps. Otherwise the step retains $c = 1$. The returned proposal is checked in the full ledger objective as specified in Appendix B.4.